STAT3003: Survey Methods
4. Cluster Sampling

Contents

*includes some unexaminable material

4.1 Introduction

- So far we have covered 2 random sample survey designs
  - Simple random sample
  - Stratified random sample
- Today we will talk about a third
  - Cluster sampling

4.1 Introduction

- Example
  - You want to estimate the average income per household in Tai Po
    - To perform a simple random sample, we would need a list (frame) of all the households (flats, houses etc.) in Tai Po. Can we find such a list?
    - We encounter the same problem with stratified random sampling – we would need a frame for each stratum
    - What can we do?

4.1 Introduction

- Example
  - Say there is actually a list of all the households in Tai Po available
    - This means we could use a simple random sample. But the households could be scattered all over Tai Po, requiring a lot of travel and time to conduct the survey
    - Stratified sampling would reduce this cost but still, it might be too high
    - What can we do?

4.1 Introduction

- A solution to both problems is the following
  - Divide Tai Po into regions (like postal codes, or blocks) then select a simple random sample of regions
  - Visit each selected region and survey every household within it
- Thus we do not need a frame of households and potentially the costs of conducting the survey are reduced, as we visit households within a region

4.1 Introduction

- This method is called cluster sampling. A formal definition is given by
  - *A cluster sample is a probability sample in which each sampling unit is a collection, or cluster, of elements*
  - (recall that sampling units are *non-overlapping collections of elements from the population that cover the entire population*)

4.1 Introduction

- Examples
  - City blocks
  - Postal codes
  - University halls/colleges
  - Orange trees (for oranges)
  - A plot in a field
  - Etc.

4.1 Introduction

- We would use cluster sampling when
  - We can't obtain a good frame of population elements at a reasonable cost, but we can obtain a frame of clusters
  - The cost of making observations increases as the distance between elements increases
- Notice that the main motivations for using cluster sampling are for keeping the cost of your survey down. It does not necessarily improve the accuracy

4.1 Introduction

- Example – Per-capita income survey
  - A sociologist wants to estimate the per-capita income in a small city. No list of resident adults is available. How should he design the sample survey?
  - Because there are no lists of elements available, cluster sampling is a good choice
  - The sociologist divides the city into clusters by dividing a map of the city into rectangular blocks, except for two industrial areas and three parks, which contain few homes

4.1 Introduction

- He decides that each city block will be a cluster, the two industrial areas will together be one cluster, as will the three parks
- He numbers each cluster, from 1 to 415
- He has calculated that he has enough resources for $n = 25$ clusters in his survey, providing he interviews every household within each cluster
- Thus he randomly selects 25 numbers between 1 and 415, according to SRS
- And the clusters with these numbers will be visited, and all the households within interviewed

4.1 Introduction

- How should we choose clusters?
  - If the measurements within a cluster are likely to be very similar (highly correlated), there is not much point in having a large cluster
  - If the measurements within a cluster are likely to be very different, then sampling a few large clusters would be beneficial
  - Often we will have to compromise between what we want statistically and what we want economically

4.1 Introduction

- What are the differences between stratified and cluster sample design?
  - ?
  - ?
  - ?

4.1 Introduction

- What are the differences between stratified and cluster sample design?
  - We sample from all the strata, but only some of the clusters
  - We take measurements from only some elements in the stratum (we perform a simple random sample in each stratum) but we measure every element in the cluster
  - The difference between strata should be large and the differences within small. The differences within each cluster should reflect the whole population, but the differences between should be small

4.1 Introduction

- Once we have decided on our clusters we need to
  - Construct a frame of clusters
  - Decide how many you want to sample
  - Choose which to sample through a SRS
  - Visit each cluster selected and measure each element within
  - Then use your data to estimate parameters…

4.2 Estimating τ

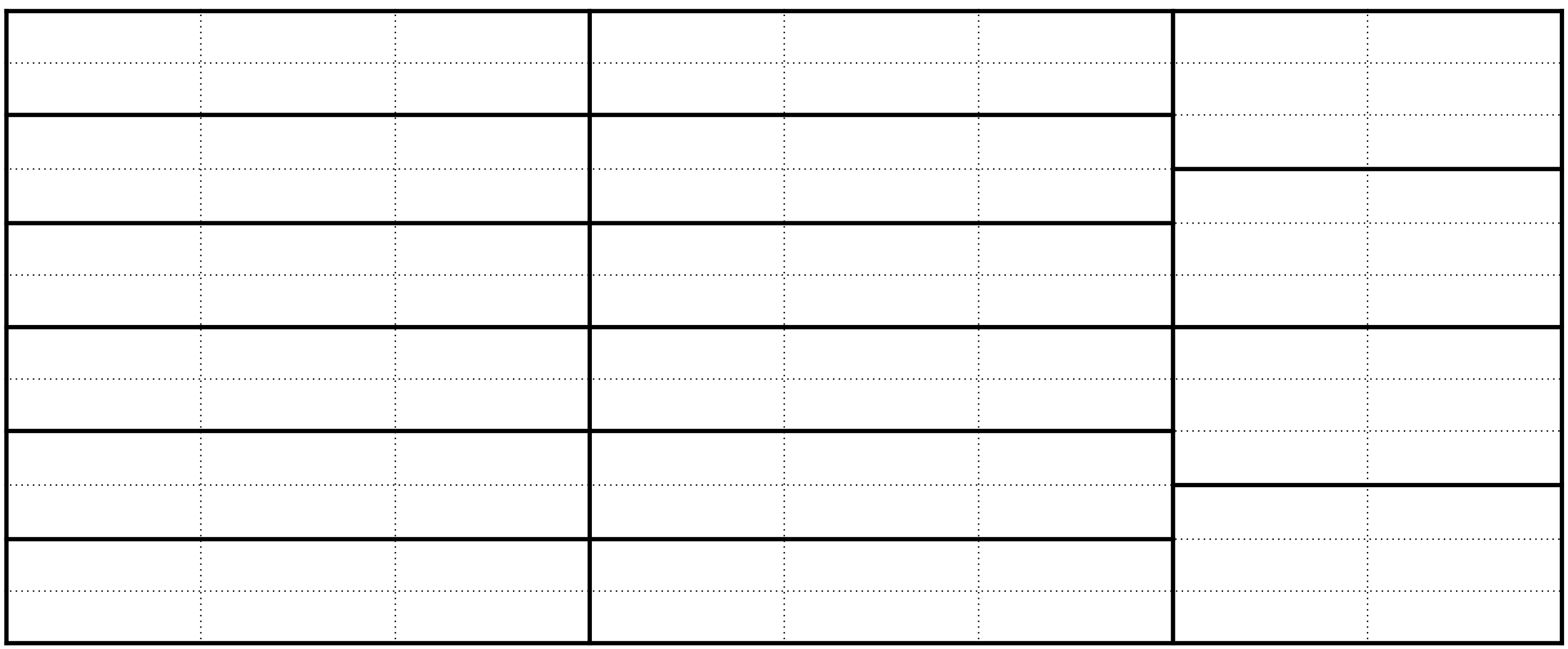
- We need some new notation. We will follow the textbook (Scheaffer et al. (2012))
  - N is the number of clusters in the population
  - n is the number of clusters selected in the sample
  - M_i is the number of elements in cluster i
  - M is the population size $(= \sum_{i=1}^{N} M_i)$
  - m_i is the sample size in the ith cluster
  - $\bar{M} = M/N$ is the average cluster size
  - Y_{ij} = the j-th observation from the i-th cluster
  - $Y_i = \sum_{j=1}^{M_i} y_{ij}$ is total of observations in the i-th cluster

4.2 Estimating τ

- Here is a population with $M = 96$ elements

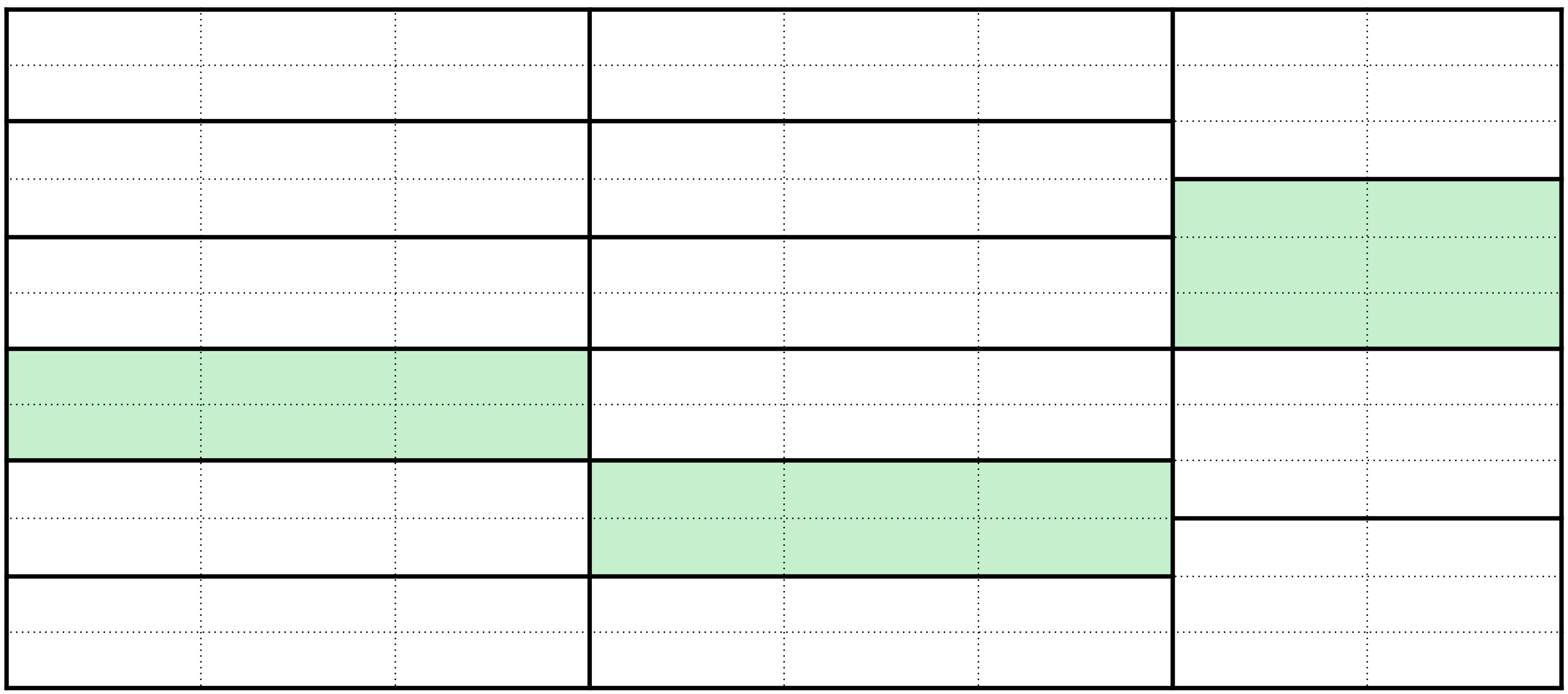
4.2 Estimating τ

- We partition the population into $N = 16$ clusters
- Each of size $m_1 = m_2 = \cdots = m_{16} = \bar{M} = 6$



4.2 Estimating τ

- We randomly select (by SRS) a sample of $n = 3$ clusters, and visit/measure every element inside



4.2 Estimating τ

- The population total is
  - $\tau = \sum_{i=1}^{N} \sum_{j=1}^{M_i} y_{ij} = \sum_{i=1}^{N} y_i$
- Consider the following estimator for total τ
  - $\hat{\tau} = N \frac{1}{n} \sum_{i=1}^{n} Y_i = N\bar{Y}$
  - This estimator works out the average total for a cluster in the sample, then multiplies it by N, because there are N clusters
- Let μ_c be the average total for a cluster
  - That is, $\mu_c = \frac{1}{N} \sum_{i=1}^{N} y_i$

4.2 Estimating τ

- To check the properties of $\hat{\tau}$, just remember it is an SRS of the cluster totals, Y_i, $i = 1, \dots, N$
- Let's check to see if it is unbiased
  - $E[\hat{\tau}] = NE[\bar{Y}] = N\mu_c$ by our results from SRS
  - $\Rightarrow E[\hat{\tau}] = N\frac{1}{N}\sum_{i=1}^{N} y_i = \sum_{i=1}^{N} y_i = \tau$
  - So $\hat{\tau}$ is unbiased.

4.2 Estimating τ

- By referring to our SRS formulae we find
  - $Var(\hat{\tau}) = N^2 Var(\bar{Y}) = N^2 \frac{N-n}{N-1}\frac{1}{n}\ \sigma_c^2$ where σ_c^2 is the variance of the cluster totals
- Hence an unbiased estimator for $Var(\hat{\tau})$ is
  - $\widehat{Var}(\hat{\tau}) = N^2\left(1-\frac{n}{N}\right)\frac{\hat{\sigma}_c^2}{n}$, where $\hat{\sigma}_c^2$ is the sample variance of the cluster totals
- Notice that, because *every* element inside a cluster is observed, the variance *within* a cluster does not appear in the above formulae.

4.2 Estimating τ

- We've seen that cluster sampling is simply an SRS of cluster totals, thus appropriate $100(1-\alpha)\%$ C.I.s are given by
  - $\hat{\tau} \pm t_{n-1,1-\frac{\alpha}{2}}\sqrt{\widehat{Var}(\hat{\tau})}$
- We may find an appropriate number of clusters to sample to achieve a $100(1-\alpha)\%$ C.I. of width $2d$ by solving
  - $d = z_{1-\frac{\alpha}{2}}\sqrt{Var(\hat{\tau})}$ for n
  - Note this will depend on σ_c^2, which is almost certainly unknown. Plug in a reasonable guess for σ_c^2 to obtain a number.

4.2 Estimating τ

- We can work out the appropriate formulae for cluster sample estimates of μ and p, again by referring to the SRS formulae
- We see that in order to have estimators with low variance, we should
  - Form clusters so that one cluster is similar to another
  - (but that is not the same as forming clusters such that the units inside a cluster are similar)
  - Form clusters which are representative of the population
- In practice, cluster sampling is used out of necessity, for reasons of economy

4.2 Estimating τ

- Usually it will give us a higher variance, but that is the price we pay for convenience or practicality
- For example, cluster sampling may be useful when
  - We can easily obtain a good frame of clusters, but not of population elements, e.g. we could easily create a frame of floors of blocks in an estate, but cannot find a frame of all estate residents
  - The cost (time, money or both) of making observations becomes prohibitive as the distance between elements grows

4.2 Estimating τ

- Example – per capita income survey
  - Interviews are conducted in each of the 25 blocks sampled, with the data shown in the table below
  - Use the data to estimate the total per-capita income in the city, and estimate the variance of your estimate

Cluster	No. residents, m_i	Total income per cluster, y_i ($000's)	Cluster	No. residents, m_i	Total income per cluster, y_i ($000's)
1	8	96	14	10	49
2	12	121	15	9	53
3	4	42	16	3	50
4	5	65	17	6	32
5	6	52	18	5	22
6	6	40	19	5	45
7	7	75	20	4	37
8	5	65	21	6	51
9	8	45	22	8	30
10	3	50	23	7	39
11	2	85	24	3	47
12	6	43	25	8	41
13	5	54			

4.2 Estimating τ

- Using the formula $\hat{\tau} \coloneqq N\bar{Y} = \left(\frac{N}{n}\right)\sum_{i=1}^{n} Y_i$, we get
  - $\hat{\tau} = \frac{415}{25} \times 1329000 = 22061400$
- To find $\widehat{Var}(\hat{\tau})$, we first calculate the sample variance of the cluster totals
  - $\hat{\sigma}_c{}^2 = \frac{\sum_{i=1}^{25}(Y_i - \bar{Y})^2}{25-1} = 21784^2$
  - Which allows us to calculate
    - $\widehat{Var}(N\bar{Y}) = N^2\left(1 - \frac{n}{N}\right)\left(\frac{\hat{\sigma}_c^2}{n}\right)$
    - $= 415^2\left(1 - \frac{25}{415}\right)\frac{21784^2}{25} = 1752766^2$

4.2 Estimating τ

- To calculate a 95% C.I.,
  - we need to look up $t_{24,0.975} = 2.06$
- Hence our 95% C.I. for the total income of the city is
  - $22061400 \pm 2.06 \times 1752766$
  - $(\$18450702, \$25672098)$ or $(\$18.5M, \$25.7M)$

4.2 Estimating τ

- Use the data to estimate μ, the mean income per capita in the city, and provide an estimate of the variance of your estimate.
- (It will help to know that the city has 2500 residents.)

4.3 Two-stage Cluster Sampling

- We often choose clusters for their convenience
- But sometimes they may contain many elements, and we cannot afford to measure each one (as required in cluster sampling)
- Or, the cluster is so homogeneous within that measuring several elements in the cluster is a waste of time

4.3 Two-stage Cluster Sampling

- Example
  - A survey wants to investigate the opinions of Hong Kong high school students about Tottenham Hotspur F.C.
    - They have a frame of high schools, from which they select a sample using simple random sampling
    - The high schools act as clusters. Each one contains 100’s of students, too many for the researchers to visit
    - The researchers decide to select a simple random sample of students from each cluster

4.3 Two-stage Cluster Sampling

- Example
  - A food inspector checks whether frozen food really does contain beef (and not horse…)
    - She samples batches of the frozen foods from the factory
    - Since each batch is likely to be very homogeneous, instead of inspecting each packet inside sample batch (cluster), she selects a simple random sample for each batch and inspects that

4.3 Two-stage Cluster Sampling

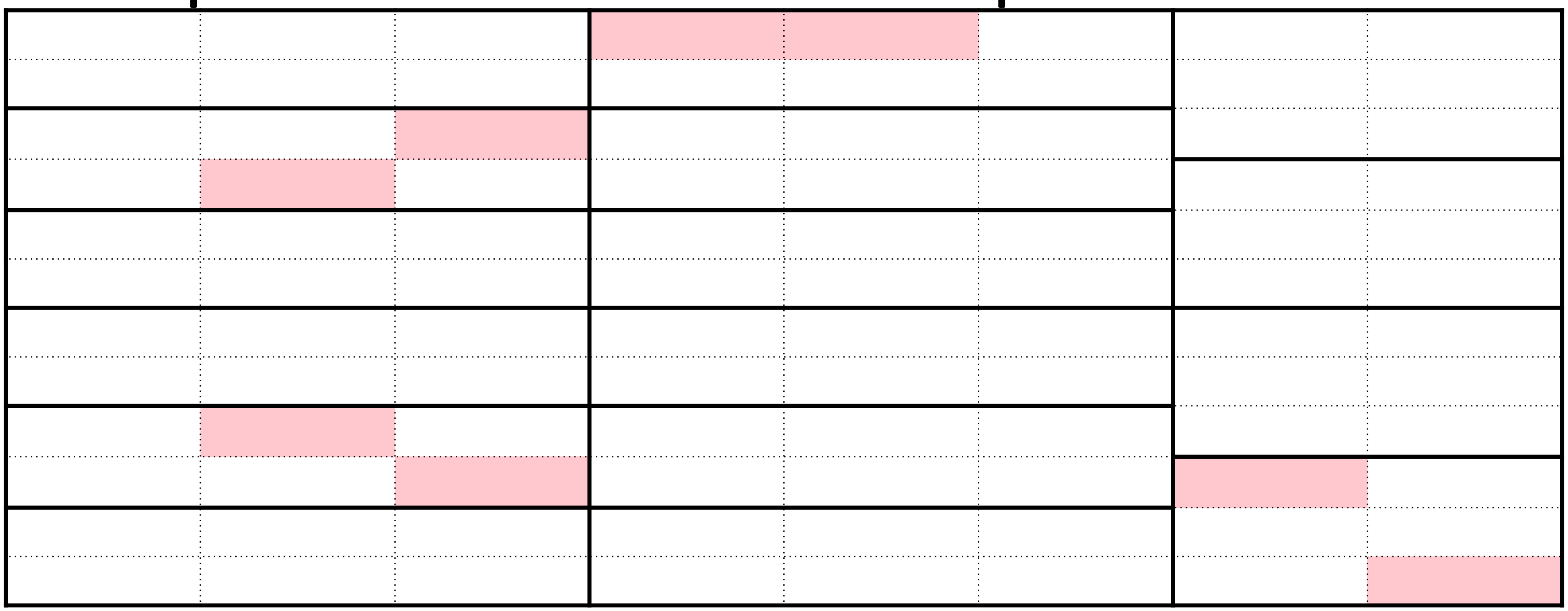
- In either case, instead of measuring every element in a cluster, we could instead randomly select elements in the cluster and measure them
- This is called two-stage cluster sampling
- *A two-stage cluster sample is obtained by first selecting a probability sample of clusters and then selecting a probability sample of elements from each sampled cluster*

4.3 Two-stage Cluster Sampling

- Two-stage cluster sampling is therefore closer to stratified random sampling than cluster sampling
- But in stratified random sampling, all the strata are sampled. With two-stage cluster sampling, only some are
- For the time being we will only consider simple random samples to select clusters and elements

4.3 Two-stage Cluster Sampling

- Below we see a population partitioned into 16 clusters of equal size. The first SRS selected 4 clusters. In the second stage, SRSs of size 2 were performed in each sampled cluster.



4.3 Two-stage Cluster Sampling

- The advantages of two-stage cluster sampling over other survey designs are the same as ordinary cluster sampling
  - More practical when a frame of all elements is not available
  - Reduces the cost of the survey if sampled elements are spread over a large geographic area

4.3 Two-stage Cluster Sampling

- How do we perform a two-stage cluster sample?
- We must decide on our clusters first
  - Ideally, we want
    - elements in each cluster to be geographically close
    - cluster sizes that are easy to administer
  - Do we want to sample a few clusters, but with many elements in each?
  - Or do we want to sample many clusters, but with a few elements in each?

4.3 Two-stage Cluster Sampling

- Large clusters tend to be heterogeneous (think city precincts), requiring large samples in each to estimate population parameters
- Small clusters tend to be homogeneous (think apartment blocks), requiring only small samples in each but many clusters over all

- Example
  - Recall the survey of high school students about Tottenham Hotspur F.C.
  - It is likely that opinions about the club would vary greatly within each school
  - Then the survey should have many representatives from each of a few schools

4.3 Two-stage Cluster Sampling

- Example
  - Imagine a survey of Hong Kong high school students about religious attitudes
  - It is likely that students within a school would hold similar opinions about religion
  - But these opinions might differ greatly between high schools
  - In this situation, we should prefer a survey of many schools, each providing a few students to the sample

4.3 Two-stage Cluster Sampling

- Usually the choice comes down to cost and what frames of clusters are available
- Once we have decided on our clusters we need to
  - Construct a frame of all the clusters
  - Draw a simple random sample of clusters
  - Construct frames of all the elements in the selected clusters
  - Draw simple random samples of elements from each of these frames

4.3 Two-stage Cluster Sampling

- The notation is the same as before for (one-stage) cluster sampling
  - N = number of clusters in population
  - n = number of clusters selected in sample
  - M = population size
  - Etc.
- But now
  - m_i = number of elements selected from ith cluster by SRS
  - Y_{ij} = the jth observation in the sample from cluster i
- Hence, $\bar{Y}_i = \frac{1}{m_i}\sum_{j=1}^{m_i} Y_{ij}$ is the sample mean for cluster i

4.3 Two-stage Cluster Sampling

- Recall that in one-stage cluster sampling, Y_i denotes the total of the observation in cluster i
- A SRS unbiased estimator for these Y_is would be $\hat{Y}_i = M_i \frac{1}{m_i} \sum_{j=1}^{m_i} Y_{ij}$
- And we suspect therefore that an unbiased estimator of $\tau = \sum_{i=1}^{N} y_i$ would be
  - $\hat{\tau} = N \frac{1}{n} \sum_{i=1}^{n} \hat{Y}_i$

4.3 Two-stage Cluster Sampling

- Two-stage cluster sampling with SRS at each stage is an example of a multistage survey design
- In order to investigate the properties of estimators in these designs, we need two results about conditional expectations and conditional variances
- We also use the terminology "primary" and "secondary" (sampling) units
  - First we sample the primary units. Let the set of primary units randomly chosen to be in the sample be s_1
  - Then, for each primary unit in s_1, we sample the secondary units inside them

4.3 Two-stage Cluster Sampling

- For two-stage cluster sampling, the primary units are the clusters and the secondary units are the elements
- For our survey about high school students and their opinion of Tottenham Hotspur
  - Primary units = high schools
  - Secondary units = high school students
- For the food inspector's survey
  - Primary units = batches of packets of frozen food
  - Secondary units = packets of frozen food

4.3 Two-stage Cluster Sampling

- There are three-stage designs. For example, household surveys conducted by the Australian Bureau of Statistics
  - Urban regions are divided into collection districts
  - Collection districtions are divided into blocks
  - Blocks are divided into dwellings
  - Therefore
    - Primary units = collection districts
    - Secondary units = blocks
    - Tertiary units = dwellings

4.3 Two-stage Cluster Sampling

- No matter the sampling design used at each stage, to investigate the properties of our estimators we need 2 results about
  - Conditional expectations
  - Conditional variances
- Let's stay with two-stage designs and let $\widehat{\theta}$ be some estimator. Recall s_1, the set of primary units chosen in the sample, is random, as is $\widehat{\theta}$
- Then
  - $E\left[E\left[\hat{\theta}\middle|s_1\right]\right] = E\left[\hat{\theta}\right]$

4.3 Two-stage Cluster Sampling

- $E\left[\hat{\theta}\middle|s_1\right]$
  – is the conditional expectation of the estimator given the selection of primary units
  – The expectation is taken over all possible subselections of secondary units from those primary units
- $E\left[\hat{\theta}\right]$
  – Is the unconditional expected value, obtained by considering all possible sample of primary units

4.3 Two-stage Cluster Sampling

- For our estimator $\hat{\tau}$
  - $E[\hat{\tau}|s_1] = N\frac{1}{n}\sum_{i=1}^{n} E[\hat{y}_i|s_1]$
  - Since the $\hat{y}_i$ are SRS estimators of the cluster totals y_i, $E[\hat{y}_i|s_1] = Y_i$
  - $\Rightarrow E[\hat{\tau}|s_1] = N\frac{1}{n}\sum_{i=1}^{n} Y_i$
- Hence
  - $E[\hat{\tau}] = E\big[E[\hat{\tau}|s_1]\big] = E\left[N\frac{1}{n}\sum_{i=1}^{n} Y_i\right] = NE[\bar{Y}]$
  - $\Rightarrow E[\hat{\tau}] = \tau$, so our estimator is unbiased.

4.3 Two-stage Cluster Sampling*

- The second result we need is
  - $Var(\hat{\theta}) = Var(E[\hat{\theta}|s_1]) + E[Var(\hat{\theta}|s_1)]$
- $Var(E[\hat{\theta}|s_1])$
  - Is the between-primary-unit variance, taken over all possible samples of primary units
- $E[Var(\hat{\theta}|s_1)]$
  - Is the expected value of the within-primary-unit variance
  - The conditional variance is obtained by considering all possible subsamples of secondary units given in the selected primary units

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4.3 Two-stage Cluster Sampling*

- To find the variance of our estimator, use
  - $Var(\hat{\tau}) = Var(E[\hat{\tau}|s_1]) + E[Var(\hat{\tau}|s_1)]$
- Looking at the first component
  - $Var(E[\hat{\tau}|s_1]) = Var\left(\frac{N}{n}\sum_{i=1}^{n} Y_i\right) = \frac{N-n}{N-1}\frac{N^2}{n}\ \sigma_c^2,$ using the formula from SRS
  - where σ_c^2 is population variance of the cluster totals
- Now consider the second, $E[Var(\hat{\tau}|s_1)]$

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4.3 Two-stage Cluster Sampling*

- We start by noticing
  - $Var(\hat{y}_i|s_1) = M_i^2 \frac{(M_i - m_i)}{M_i - 1}\frac{\sigma_i^2}{m_i}$
  - Since $\hat{y}_i$ is the estimator of cluster total y_i under SRS within cluster i
- Hence, it follows
  - $Var(\hat{\tau}|s_1) = Var\left(\frac{N}{n}\sum_{i=1}^{n}\hat{y}_i \,\middle|\, s_1\right) = \frac{N^2}{n^2}\sum_{i=1}^{n} Var(\hat{y}_i|s_1)$
  - Because the primary units are chosen independently
  - $\Rightarrow Var(\hat{\tau}|s_1) = \frac{N^2}{n^2}\sum_{i=1}^{n} M_i^2 \frac{(M_i - m_i)}{M_i - 1}\frac{\sigma_i^2}{m_i}$

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4.3 Two-stage Cluster Sampling*

- All that is left is to find $E[Var(\hat{\tau}|s_1)]$
  - This is the expected value of the conditional variance, over all possible samples of clusters
- We use a familiar trick
  - Let $Z_i = 1$ if cluster i is in the sample, 0 otherwise
  - $\Rightarrow Var(\hat{\tau}|s_1) = \frac{N^2}{n^2} \sum_{i=1}^{N} M_i^2 \frac{(M_i - m_i)}{M_i - 1} \frac{\sigma_i^2}{m_i} Z_i$
  - $\Rightarrow E[Var(\hat{\tau}|s_1)] = \frac{N^2}{n^2} \sum_{i=1}^{N} M_i^2 \frac{(M_i - m_i)}{M_i - 1} \frac{\sigma_i^2}{m_i} \frac{n}{N}$

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4.3 Two-stage Cluster Sampling

- We have already shown $\hat{\tau}$ is unbiased
- Furthermore, we can show
  - $Var(\hat{\tau}) = N^2 \frac{N-n}{N-1} \frac{\sigma_c^2}{n} + \frac{N}{n} \sum_{i=1}^{N} M_i^2 \frac{M_i - m_i}{M_i - 1} \frac{\sigma_i^2}{m_i},$
  - where σ_c^2 is the population variance of the cluster totals, i.e. the y_is and σ_i^2 is the population variance within the ith cluster
  - $\widehat{Var}(\hat{\tau}) = N(N-n) \frac{1}{n} \hat{\sigma}_c^2 + \frac{N}{n} \sum_{i=1}^{n} M_i (M_i - m_i) \frac{1}{m_i} \hat{\sigma}_i^2$
  - where $\hat{\sigma}_c^2$ is the sample variance of the estimated cluster totals (the $\hat{Y}_i$)
  - and $\hat{\sigma}_i^2$ is the sample variance inside cluster i

4.3 Two-stage Cluster Sampling

- When it comes to calculating a $100(1-\alpha)\%$ C.I. for τ, which percentile should we use?
  - Given the complicated form of $\widehat{Var}(\hat{\tau})$, it is common practice to simply use $z_{1-\frac{\alpha}{2}}$ rather than a percentile from the t distribution
  - For large sample sizes this is reasonable.
  - For small to medium sample sizes, or if you are concerned about the normality of your estimator, calculate your C.I. and note you have made this approximation
- Thus our $100(1-\alpha)\%$ C.I. for τ will be
  - $\hat{\tau} \pm z_{1-\frac{\alpha}{2}}\sqrt{\widehat{Var}(\hat{\tau})}$

4.3 Two-stage Cluster Sampling

- Example – clothes manufacturer (μ)
  - A clothes manufacturer has 90 factories located across China
  - She wants to estimate the average number of hours that the sewing machines were down for repairs over the past six months
  - The plants are scattered all over China, and each factory contains many machines, so she decides to use a two-stage sampling design
  - Her resources allow her to sample $n = 10$ factories and roughly 20% of the machines in each factory
  - She knows that her company has 4500 machines in total

4.3 Two-stage Cluster Sampling

Factory	M_i	m_i	$\bar{y}_i$	s_i
1	50	10	5.4	3.37
2	65	13	4	3.27
3	45	9	5.666666667	4.09
4	48	10	4.8	3.65
5	52	10	4.3	3.33
6	58	12	3.833333333	3.86
7	42	8	5	2.27
8	66	13	3.846153846	2.08
9	40	8	4.875	2.47
10	56	11	5	3.44

4.3 Two-stage Cluster Sampling

- Note we are estimating μ, not τ. We must adjust our formulae
  - Since $\mu = \frac{\tau}{M}$, take $\hat{\mu} = \frac{\hat{\tau}}{M}$. Then $\widehat{Var}(\hat{\mu}) = \frac{1}{M^2}\,\widehat{Var}(\hat{\tau})$.
- We estimate the average downtime per machine by $\hat{\mu}$
  - $\hat{\mu} = \left(\frac{N}{M}\right)\frac{1}{n}\sum_{i=1}^{n} M_i\bar{y}_i = \left(\frac{90}{4500}\right)\left(\frac{1}{10}\right)\sum_{i=1}^{10} M_i\bar{y}_i = 4.80$
- We estimate the variance of $\hat{\mu}$ using $\widehat{Var}(\hat{\mu})$
  - $\widehat{Var}(\hat{\mu}) = \frac{1}{M^2}N(N-n)\frac{1}{n}\hat{\sigma}_c^2 + \frac{1}{M^2}\frac{N}{n}\sum_{i=1}^{n} M_i(M_i - m_i)\frac{1}{m_i}\hat{\sigma}_i^2$
  - $\Rightarrow = \frac{1}{4500^2}90 \times \frac{80}{10}s_c^2 + \frac{1}{4500^2}\frac{90}{10}\sum_{i=1}^{10} M_i(M_i - m_i)\frac{1}{m_i}s_i^2$

4.3 Two-stage Cluster Sampling

- The s_i^2's are simply the sample variances from cluster i (already calculated)
- Plugging in the numbers, we get $s_c^2 = 27.72^2$
- Hence, the estimated variance is given by
  - $\widehat{Var}(\hat{\mu}) = \frac{8}{90\times 50^2}(27.72^2) + \frac{1}{10\times 90\times 50^2}(21985)$
  - $\Rightarrow \widehat{Var}(\hat{\mu}) = 0.0371$
- With an estimate of the variance, we can provide a bound on the error of estimation (an approximate 95% CI) of
  - $\hat{\mu} \pm 1.96\sqrt{\hat{V}(\hat{\mu})} = 4.80 \pm 1.96\sqrt{0.0371} = 4.80 \pm 0.38$

4.3 Two-stage Cluster Sampling

- The formulae for estimating p follow quickly from the formulae for μ
  - Remember that p is itself a population mean
- Hence an unbiased estimator is
  - $\hat{p} = \frac{N}{M}\frac{1}{n}\sum_{i=1}^{n}\hat{Y}_i$, where $\hat{Y}_i$ is the estimator for the number of people in cluster with the characteristic
  - We may write $\hat{Y}_i = M_i\hat{p}_i$ to relate p to the sample proportions inside the clusters

4.3 Two-stage Cluster Sampling

- And an unbiased estimator for the variance is
  - $\widehat{Var}(\hat{p}) = \frac{1}{M^2} N(N-n) \frac{1}{n} \hat{\sigma}_c^2 + \frac{1}{M^2} \frac{N}{n} \sum_{i=1}^{n} M_i (M_i - m_i) \frac{1}{m_i} \hat{\sigma}_i^2$
- Recall
  - $\hat{\sigma}_i^2$ is the estimator for population variance of cluster i, hence $\hat{\sigma}_i^2 = \frac{m_i}{m_i - 1} \hat{p}_i (1 - \hat{p}_i)$
  - $\hat{\sigma}_c^2$ is the sample variance of the estimated cluster totals, hence
    - $\hat{\sigma}_c^2 = \frac{1}{n-1} \sum_{i=1}^{n} \left(M_i \hat{p}_i - \frac{M}{N} \hat{p} \right)^2$

4.3 Two-stage Cluster Sampling

- Example – clothes manufacturer (p)
  - The clothes manufacturer wants to estimate the proportion of machines that have been shut down for major repairs
  - This table gives the sample proportions

Factory	M_i	m_i	$\hat{p}_i$
1	50	10	0.40
2	65	13	0.38
3	45	9	0.22
4	48	10	0.30
5	52	10	0.50
6	58	12	0.25
7	42	8	0.38
8	66	13	0.31
9	40	8	0.25
10	56	11	0.36

4.3 Two-stage Cluster Sampling

- Our point estimate is given by
  - $\hat{p} = \left(\frac{N}{M}\right)\frac{1}{n}\sum_{i=1}^{n} M_i\hat{p}_i = \frac{90}{4500}\left(\frac{1}{10}\right)\sum_{i=1}^{10} M_i\hat{p}_i = 0.352$
- And estimate for the variance is given by
  - $\widehat{Var}(\hat{p}) = \frac{1}{M^2}N(N-n)\frac{1}{n}\hat{\sigma}_c^2 + \frac{1}{M^2}\frac{N}{n}\sum_{i=1}^{n} M_i(M_i - m_i)\frac{1}{m_i - 1}\hat{p}_i(1-\hat{p}_i)$
  - Where $\hat{\sigma}_c^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(M_i\hat{p}_i - \frac{M}{N}\hat{p}\right)^2$
  - Plugging in the numbers, we find $\widehat{Var}(\hat{p}) = 0.037^2$
- Hence our approximate 95% CI is
  - 0.352 ± 0.073